

EXERCISE 5

REVERSE (SINGLE – DOUBLE)

THEORETICAL PART

Purpose: Understanding the operating principle of rectifier circuits. Study of the basic rectifier circuits of alternating (sinusoidal) voltage. Finally, application of such circuits.

In all electronic devices that use transistors or integrated circuits that need constant voltages which are usually provided by the mains voltage (220 Volt, 50 Hz, alternating) with some kind of rectification. The main component for the rectification is the silicon or germanium diode which allows electrons to circulate through it always from the cathode to the anode and even when the anode becomes positive in relation to the cathode.

We will first examine the simple resistance rectification. We apply to sinusoidal voltage V_i the ends A and B of the parallel circuit in Fig.1.

During the positive oscillations of V_i the anode of diode d becomes positive with respect to the cathode and current flows through the rectifier. During the negative oscillations no current flows because the anode becomes negative with respect to the cathode.

The current flowing through the positive oscillations causes a voltage drop V_0 across the resistor R. Since the changes in current follow the changes in voltage, the output voltage V_0 will vary as the positive oscillations

of the input voltage V_i , which caused the current. Figure 1b shows the waveforms of the input and output voltages V_i and V_0 as a function of time and in time correspondence.

The voltage V_0 is now a continuous voltage in the form of pulses. The amplitude of the voltage V_0 is slightly smaller than the amplitude of the voltage V_i because when the rectifier is conducting

a voltage drop is created within it. In general, the internal resistance of the rectifier must be as small as possible so that the highest possible voltage V_0 remains at the output, at the ends of R.

Another way of rectification is carried out by the parallel circuit of figure 2a.

In this case, the following happens:

In the positive half-cycle of V_i the diode D it is reverse biased so it does not conduct, so it behaves like a switch S1. So at points C and D V_i is transferred and we have $V_i = V_0$ from 0 to $T/2$.
2. In the negative half-cycle the diode conducts because we have forward polarization of the diode. So S1 is closed with the result that the ends C and D are short-circuited and we have a potential difference of 0.6 Volt, in the time period from $T/2$ to T .

In double or full rectification we will examine the following two circuits: those of Figures 3a and 4a. In the adjacent Figure 3a the

transformer in the secondary winding between points C and D has a medium tap. That is, the number of turns from C to H is equal to the number of turns from H to D, and therefore the voltages in the two sections of the secondary are equal. In the circuit there are two diodes D1 and D2 which are alternately conductive because the voltages prevailing in the two anodes C and D are in a phase difference of 180°

between them with respect to the common point H and therefore when one diode is conductive the other is not and vice versa.

Here we note that the amplitude of the voltage V_0 is half the secondary voltage at points C, D. The input and output waveforms in time-varying units of the circuit of Fig. 3a are shown in Fig. 3b.

A second double or full rectifier circuit is the bridge rectifier circuit, shown in Figure 4a. The circuit consists of four diodes in a bridge configuration. This circuit has the most advantages over the circuit shown in Figure 3a. Also, in these rectifier circuits, the use of silicon diodes has become widespread because they have very low internal resistance in the forward direction of the current.

Its operation is as follows:
During the positive transition (Fig. 4b) of the sinusoidal input voltage V_i , point A is positive with respect to B. Then the anode of D1 is positive with respect to its cathode, i.e. the crystal diode receives a forward voltage and becomes conductive. The cathode of D3 connected to point B

is negative with respect to its rise, so D3 also becomes conductive. However, it is clear that D2 and D4 are reverse biased during the positive switching and therefore are not conductive. Also in figure 4b we see that the current flows only through diodes D1 and D3.

In Fig. 4c it is seen that during the negative switching when point A is negative with respect to B, the anode of D2 is positive with respect to its cathode and the cathode of D4 is negative with respect to the anode. Therefore diodes D2 and D4 are now 33

forward biased, while D1 and D3 are reverse biased. Therefore D2 and D4 allow current to pass through R. The polarity of the voltage developed across R is the same as when D1 and D3 were forward biased. Thus, diode D1 in series with D3 rectifies the positive alternation of the sinusoidal input, while D2 in series with D4 rectifies the negative alternation.

In the adjacent figure 4d, the input-output waveforms (double rectification) of the bridge circuit of figure 4a are shown in time correspondence.

Fig. 4d

Therefore, the bridge circuit 4a is a double-rectifier circuit and presents the following similarities rectifier circuit with and differences (advantages-disadvantages) when compared to the usual double-a medium-tap transformer of Fig. 3a:

- a) In both circuits Fig. 3a (rectification with MT of medium reception) and Fig. 4a (bridge) the output frequency F_{out} is twice the input frequency F_{in}
- b) Double-inverting bridge circuits do not require a double winding on the secondary of the transformer.
- c) In the bridge the input voltage V_i is equal to the output voltage $V_0 / 2$, while in contrast in the double rectification of the circuit of Fig. 3a the voltage V_0 is equal to $V / 2$.
- d) A disadvantage of the bridge rectifier circuit is that at each switching the DC current of the circuit must pass through two diodes in series. Consequently, the voltage drop during the conducting switching across these two diodes will be greater than the voltage drop across a single diode.

LABORATORY PART

1. a. Make the circuit of figure 1a. apply an alternating voltage of 8 Vpp and frequency of 1 KHz and observe and draw the output waveform in correspondence with the input. **(Note: For all the waveforms that you will draw in ALL paragraphs, the measurements on the axes should be noted and the oscilloscope should have DC pressed on both channels.)**

- b. Using the oscilloscope, measure V_{max} and with the voltmeter in the DC position, measure the average value ($V_{\bar{y}}$) of the output voltage. Also valid: $V_{\bar{y}} = V_{max} / \pi$ calculate the average value and compare it with the measured one.

V_{max} (V)		$V_{\bar{y}}$ (V)		$V_{\bar{y}}$ (V) =	
Pulse rate		Voltmeter		V_{max} / π	

Table 1

c. Press the X – Y button of the oscilloscope and draw the transfer characteristic

2. a. Make the circuit of figure 2a and repeat the task of paragraph 1.

V _{max} (V)		V _ȳ (V)		V _ȳ (V) =	
Pulse rate		Voltmeter		V _{max} / p	

Table 2

b. Also change the polarity of diode D in figure 2a and repeat the task in paragraph 1. What do you observe?

V _{max} (V)		V _ȳ (V)		V _ȳ (V) =	
Pulse rate		Voltmeter		V _{max} / p	

Table 3

3. a. Make the circuit of figure 4a. Apply (supply) an alternating voltage of 8 V_{pp} and frequency of 1 KHz, observe and draw the output waveform in correspondence with the input. Here we use only one beam of the oscilloscope.

b. Measure the average value of the output voltage V_ȳ with a voltmeter in the DC position and the output frequency and V_{max} on the oscilloscope.

V _ȳ (V)		Error		V _{max} (V)		V _ȳ = 2V _{max} /ȳ	
Voltmeter		(Hz)		Pulse rate		(V)	

Table 4

c. Disconnect diode D2. Observe and draw the waveform.

QUESTIONS

1. Analysis of circuit operation and explanation of output waveforms.
2. If we placed the diode in the circuit of figure 1a upside down then what would it be? the output waveform and why?
3. What is the difference between circuits 3a and 4a and why?
4. Where are rectifier circuits used and why?
5. Implement the circuit of Figure 3a in Multisim using the TS_MISC_VIRTUAL transformer and 1N4148 or 1N4004 diodes. We supply the transformer with 220 V AC (points A – B) and with a frequency of 50 Hz (mains frequency). With a dual-beam oscilloscope, we place the A(+) channel at point C and the B(+) channel at point E. Design, measure and observe the output voltage and frequency at point E. Press the B/A button of the oscilloscope and draw the transfer characteristic.